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Effects of competition, shade and soil conditions on the recolonization of three forest herbs in tree-planted riparian zones

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Agricultural legacies; Assembly rules; Ecological restoration; Environmental filters; Recolonization limitation; Recruitment and establishment niches; Riparian zones

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Abstract

Questions: In mesic forests, ecological filters due to past agricultural land use reduce forest herb recolonization. Is the recruitment of such species also limited in tree-planted riparian zones by local filters such as competition, shade level and soil conditions?

Location: Two agricultural watersheds, southeastern Québec, Canada.

Methods: Three herbs characteristic of natural riparian forests were selected for this study: one graminoid, *Glyceria striata*, and two ferns, *Matteuccia struthiopteris* and *Onoclea sensibilis*. Effects of shade level (75% vs 50%) and soil type (forest vs agricultural soil) on seedling emergence were evaluated in a seed-sowing greenhouse experiment. In a 2-yr transplant field experiment, seedling and sporophyte establishment was monitored in five natural riparian forests and five tree-planted post-agricultural riparian zones on microsites with understorey vegetation kept intact or cleared and on forest or agricultural soils. Using *a priori* contrasts, we assessed the influence of habitat type (natural riparian forests or tree-planted riparian zones), competition and post-agricultural soil type on transplant survival and growth.

Results: Seedling emergence tended to be higher on forest soils for *G. striata* while sporophyte emergence increased under 75% shade for *M. struthiopteris*. Transplanted seedlings and sporophytes of the three species survived and grew as well in tree-planted riparian zones as in natural riparian forests. In tree-planted riparian zones however, competing understorey vegetation reduced the survival and growth of *G. striata* and agricultural soil reduced the growth of *M. struthiopteris*. For *O. sensibilis*, only sporophyte survival was reduced by competition in tree-planted riparian zones.

Conclusions: Planting trees in post-agricultural riparian zones fosters establishment of forest herbs similar than those observed in natural riparian forests. Additional environmental filters specific to tree-planted riparian zones, however, offset the positive influence of trees and limit recolonization of the three studied species. Considering the partial restoration success of establishment niches by tree planting, controlling spontaneous vegetation after tree planting is advised when conceivable and cost-effective to promote the recolonization of environmentally-limited forest herbs. Long-term transplant experiments should be more often conducted to identify the ecological filters that reduce plant recolonization, and thereby design the most effective restoration strategies.

Introduction

The recovery of forest plant diversity is a critical process in post-agricultural landscapes because of persistent effects of past land use (Dupouey et al. 2002; Flinn & Vellend 2005; Hermy & Verheyen 2007). In recent forests established on former farmlands, agricultural legacies generally result in impoverished plant communities lacking many species characteristic of ancient forests, i.e. of forests that have existed continuously since the earliest land-use maps (Hermy et al. 1999; Flinn & Vellend 2005; Hermy & Verheyen 2007). The low recolonization of forest herbs in recent forests is related to a bottleneck process structured around three successive stages of the plant life cycle, namely, the dispersal of seeds, the recruitment of seeds into seedlings and the establishment of seedlings into adult individuals able to reproduce (Verheyen & Hermy 2004; Hermy & Verheyen 2007; Cramer et al. 2008; Baeten et al. 2009a,b).

Traditionally, low forest herb recolonization in recent forests has been attributed predominantly to dispersal limitation rather than to recruitment limitation (Dzwonko & Loster 1992; Brunet & Von Oheimb 1998; Ehrlén & Eriksson 2000; Clark et al. 2007; Hermy & Verheyen 2007). The habitat fragmentation of post-agricultural landscapes indeed often prevents the seeds of forest herbs from even reaching recent forests (Clark et al. 2007; Cramer et al. 2008). The absence of dispersal vectors is furthermore a major obstacle for the recolonization of species with specific requirements, such as myrmecochorous forest herbs (i.e. species dispersed by ants; Petersen & Philip 2001; Heinken 2004). Dispersal limitation has been notably demonstrated in forest herb seed sowing experiments, which have usually found similar germination rates in recent and ancient forests (Donohue et al. 2000; Graae et al. 2004; Verheyen & Hermy 2004). The distribution patterns of ancient forest species are also generally negatively correlated with the isolation of forest patches (Peterken & Game 1984; Dzwonko 1993; Grashof-Bokdam & Geertsema 1998; Bossuyt et al. 1999; Hermy et al. 1999; Verheyen & Hermy 2001). The slow colonizing ability characterizing herbs of ancient forest has thus pointed to dispersal limitation as the major driver explaining their low recolonization in recent forests.

The degradation of local environmental conditions associated with past agricultural land use also limits the recruitment (i.e. seedling emergence and establishment) of ancient forest herbs transplanted in recent forests (Verheyen & Hermy 2004; Cramer et al. 2008; Baeten et al. 2009a,b). In these forests, soils are often characterized by higher pH and nutrient concentrations (especially total phosphate) and lower organic matter than soils of ancient forests (Bossuyt et al. 1999; Honnay et al. 1999; Verheyen et al. 1999). Such edaphic modifications generally foster the growth of opportunistic, fast-colonizing forest herbs, and thereby directly impact community composition (Fraterrigo et al. 2006; Baeten et al. 2010, 2011) and indirectly limit the recruitment of transplanted forest herbs through competition (Flinn & Vellend 2005). For example, in recent forests of central Belgium and Poland, the ruderal competitive herb *Urtica dioica* L. benefits from high phosphorus availability and out-competes forest herbs (Honnay et al. 1999; De Keersmaeker et al. 2004; Endels et al. 2004; Baeten et al. 2009a; Orczewska 2009). The species-specific response to altered soil conditions ultimately results in different species assemblages between recent and ancient forests (Fraterrigo et al. 2006). Transplant experiments are therefore a promising perspective to identify the ecological filters limiting plant recolonization by relating transplant performance to environmental factors (i.e. by using plants as phytometers; Sanders & McGraw 2005), and have greatly contributed to understand the impacts of past agricultural land use on the recolonization of forest herbs in recent mesic forests (Hermy & Verheyen 2007). However, such experiments have to date been rarely used in a restoration context (Dietrich et al. 2013), and little is known of the assembly rules for riparian tree plantations implemented for agri-environmental purposes.

In agricultural landscapes worldwide, significant effort has been deployed to mitigate the detrimental impacts of intensive agriculture on water resources through tree plantations in riparian zones (Lowrance et al. 1984; Schultz et al. 1995; Lee et al. 2000, 2003; Marshall & Moonen 2002; Boutin et al. 2003; Lovell & Sullivan 2006). As in recent mesic forests, agricultural legacies may, however, limit the recolonization of forests herbs in tree-planted riparian zones. In riparian zones, plant dispersal is fostered by rivers, which act as major ecological corridors for long-distance, water-mediated seed transport (Nilsson et al. 1991, 2010; Naiman et al. 1993; Johanson et al. 1996; Tabacchi et al. 1998; Andersson et al. 2000; Boedeltje et al. 2003; Gurnell et al. 2008), although dispersal limitation of forest herbs has sometimes been noted after tree planting in riparian zones (Battaglia et al. 2002, 2008; McClain et al. 2011). In addition to promoting plant dispersal, water flow also contributes to the destruction of existing vegetation in riparian zones (Tabacchi et al. 1998) and to the regular reorganization of soils through sediment erosion and deposition (Hupp & Osterkamp 1996; Bendix & Hupp 2000; Osterkamp & Hupp 2010). This dynamic process is likely to mitigate the impacts of competition and soil changes associated with agricultural land use on riparian plant composition. A better understanding of competition and soil effects on the recolonization of forests herbs in tree-planted riparian zones can thus be highly beneficial for the ecological restoration of riparian biodiversity.

This study aims to determine the importance of environmental limitations related to competition, shade and soil for the recolonization of three forest herbs in tree-planted riparian zones by combining seed sowing and transplant experiments. More specifically, we hypothesized that high shade level and forest soil characteristic of natural riparian forests increase seedling emergence of forests herbs compared to low shade level and agricultural soil typical of

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tree-planted riparian zones. We also hypothesized that increased plant competition and altered edaphic conditions of agricultural soils in tree-planted riparian zones reduce the establishment (survival and growth) of seedlings compared to natural riparian forests.

Methods

Study area and species

The field study was conducted in two agricultural watersheds of southeastern Québec (Canada). These watersheds are characterized by gleysolic and brunisolic soils, a mean annual temperature of 4 °C (19 °C in July and −12 °C in January) and 1300 mm mean annual precipitation, of which 24% falls as snow (Environment Canada 2015, www.ec.gc.ca). The Boyer watershed (46°41′ N, 70°55′ W) is a 216-km² area with scattered forest fragments (24% of the land) and a large expanse of agricultural land (66% of land use), of which 26% is farmed with annual crops (OBV Côte-du-Sud/GIRB 2011). The Bélair watershed (46°26′ N, 70°56′ W) covers 43 km²; 66% of which is forested, and 33% is devoted to agricultural use, of which annual crops constitute 18% (MAPAQ *unpubl data*). Trees were planted in the riparian zones of these two watersheds from 1995 to 2009, after the provincial government banned agriculture in a zone at least 3-m wide along streams (Gouvernement du Québec 1987). The six species planted most commonly were *Fraxinus pennsylvanica* Marsh., *Acer saccharum* Marsh., *Picea glauca* (Moench) Voss, *Larix laricina* (Du Roi) K. Koch, *Quercus rubra* L. and *Quercus macrocarpa* Michx.

A data set of 784 1-m^2 botanical surveys conducted in 53 tree-planted riparian zones and 14 natural riparian forests in the study area during 2012 was used to perform an indicator value analysis (Dufrêne & Legendre 1997). We then selected three herb species for this study from among the 13 that were known indicators of natural riparian forests. We selected one graminoid, *Glyceria striata*, and two fern species, *Matteuccia struthiopteris* and *Onoclea sensibilis* for our study, due to their height (tall) and high growth variation, relative ease of cultivation and availability in local nurseries.

Greenhouse seed-sowing experiment: design and monitoring

A seed-sowing experiment was conducted in greenhouse to assess the effects of two shade levels (50% and 75%) and two soil types (agricultural soil and forest soil) on seedling and sporophyte emergence of the three studied species. These abiotic parameters were selected as they reflect tree cover and soil conditions prevailing in the tree-planted riparian zones and natural riparian forests monitored in the transplant experiment (see next section). Shade levels indeed corresponded approximately (within 3% to 4%) to visual estimates of tree cover in the studied tree-planted riparian zones and natural riparian forests (Appendix S1). Similarly, agricultural and forest soils were collected in the 0–15 cm soil horizon of transplant experiment sites (see next section) for a total of *ca*. 1 000 cm^3 per site.

This seed-sowing experiment was performed using a split-plot design in which main plots corresponded to the two shade treatments and subplots to the two soil treatments. Seven replicates of main plots were established using superimposed layers of fiberglass nets to obtain the desired shade levels. These shade levels thus corresponded to a 50% and 75% reduction (representative of 54 ± 6% and 78 ± 5% cover of trees on average in tree-planted riparian zones and natural riparian zones, respectively) of the photosynthetic photon flux density in full light (50 $\mu mol \cdot m^{-2} \cdot s^{-1}$) measured with a photometer. Subplots corresponded to gardening pots (L × l × h = 8 × 8 × 9 cm) randomly filled with post-agricultural or forest soil for a total of two subplots per main plot (for each species). Prior to the experiment, both types of soil were pasteurized at 60 °C for 4 h to reduce their seed banks and kill most pathogens.

Each experimental unit was sown in Jun 2014 with either 1 g spores of *O. sensibilis*, 1 g spores of *M. struthiopteris*, or 50 seeds of *G. striata*. Spores were extracted by sieving fertile fronds of *O. sensibilis* and *M. struthiopteris* collected in the field 1 month prior to the experiment and then stored in dry and dark conditions at 4 °C, while seeds of *G. striata* were obtained from a local nursery (estimated seed viability: 75%). Greenhouse environmental conditions were maintained at 24 °C and 50% relative humidity (RH) during 15 daylight hours, alternating with nine nighttime hours at 18 °C and 70% RH. Experimental units were monitored daily and watered with rain water to keep soil moist (i.e. generally every 1–2 days). The number of sporophytes (for *O. sensibilis* and *M. struthiopteris*) and seedlings (for *G. striata*) emerging from experimental units was counted three times per week for the 14 wk of the experiment, a total of 39 counts, during which seedlings of spontaneous species were removed. Due to pasteurization, these spontaneous seedlings were scarce and also did not correspond to the three studied forest herbs and thus can be considered absent from the soil seed banks.

Field transplant experiment: design and monitoring

A transplant field experiment was conducted in the two studied watersheds to assess the effect of two riparian habitat types (riparian zones with trees of *ca*. 8–10-m high planted 15–17 yrs prior the experiment, hereafter referred to as *PLANTATION*, and natural riparian forests with trees of *ca*. 20-m high, hereafter *FOREST*) and three microsites on the seedling and sporophyte establishment of the three studied species using an incomplete factorial design (Fig. 1). Specifically, these microsite treatments were established to evaluate the effects of competition (presence–absence of spontaneous understorey vegetation) and two soil types (agricultural soil–forest soil) on the seedling survival and growth of the three studied species. *In-situ* growth conditions were represented by the first microsite treatment corresponding to *in-situ* soil with spontaneous vegetation (hereafter, microsite on *in-situ* soil with vegetation). The second microsite treatment used *in-situ* soil, but standing spontaneous vegetation and litter were removed by hand to eliminate competition (hereafter, microsite on *in-situ* soil with cleared vegetation). In the third microsite treatment, soils were exchanged between *PLANTATION* and *FOREST* to a depth of *ca*. 30 cm after living and dead spontaneous vegetation had been cleared (hereafter, microsite on *ex-situ* soil with cleared vegetation). Note that *in-situ* soils corresponded to agricultural soil in *PLANTATION* and to forest soil in *FOREST*, and *ex-situ* soils to the opposite. As vegetation was hand-weeded prior to the soil exchange, a full factorial design could not be established for this experiment, i.e. the microsite treatment on *ex-situ* soil with vegetation could not be studied.

To account for between-river hydrological variability, the two riparian habitat types were paired along five rivers (three rivers on the Boyer watershed and two on the Bélair watershed). A total of ten experimental units (or sites) was thus studied, i.e. five tree-planted riparian zones (*PLANTATION*) and five natural riparian forests (*FOREST*).

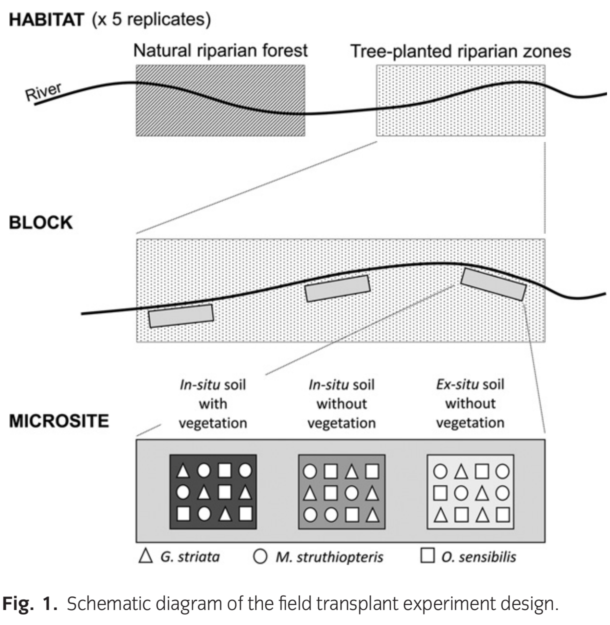

Fig. 1. Schematic diagram of the field transplant experiment design.

The studied sites were also selected for their relative homogeneity based on vegetation (covers of trees, shrubs, herbaceous, bryophytes and litter), soil (soil pH_{water}, % clay, %silt, %sand, cation exchange capacity, %organic matter, P content, Ca^{2+} saturation, K^{+} saturation, Mg^{2+} saturation) and microclimate measurements (mean daily temperature, air RH, soil RH). Among these 18 environmental variables, only five differed significantly between habitat types: the cover of trees as well as soil RH and proportion of organic matter were higher in natural riparian forests compared to tree-planted riparian zones, while the cover of shrubs and mean daily temperature were lower (Appendix S1). Furthermore, dominant tree species were also similar between *PLANTATION* and *FOREST*, i.e. *Fraxinus pennsylvanica*, *Acer saccharum* and *Quercus* spp. (see Appendix S2 for the full list of plant species).

On each site, the three microsite treatments were established in three 1 m × 2 m contiguous plots forming a block and were replicated in three blocks, for a total of nine plots per site. Four seedlings of each studied species were transplanted randomly every 20 cm in each plot at the end of Jun 2013 (1440 seedlings in total) and identified with a small plastic stake in order to relocate them easily (Fig. 1). Seeds of *G. striata* from a local nursery (Prairie Moon Nursery, Winona, MN, US) were grown for 1 month in a greenhouse to obtain seedlings, while seedlings of *O. sensibilis* and *M. struthiopteris* were provided by a local nursery (Fougères Boréales, Sainte-Sophie, Québec, CA). To reduce pre-transplanting growth effects, individuals of similar size (ca. 7–10-cm tall) were chosen when transplanting in the field.

Transplanted individuals were monitored during two consecutive growing seasons (i.e. in summers of 2013 and 2014; mean temperature from June to August = 17.5 °C in 2013 and 18.4 °C in 2014; cumulative precipitation from June to August = 330 mm in 2013 and 413 mm in 2014). Transplants were measured every 3 wks in 2013 from the week after transplanting (initial size) to the end of September, for a total of three measurement times, and every 5 wks in 2014 from the beginning of June to the end of August, for a total of three measurement times. During each monitoring, the survival (%) of each species was calculated for each microsite by counting the number of dead and living transplants. Furthermore, we assessed transplant growth using five morphological measurements. Specifically, we counted the number of stems and leaves of each transplant and measured total height (cm) and height from the ground to the first leaf (or leaflet for the two fern species; cm). We also measured total width (cm) for *O. sensibilis* and *M. struthiopteris* and the length of the longest leaf (cm) for *G. striata*. Only the final survival and measurements after two growing seasons (end of August 2014) were retained for the analysis. In microsite plots with the

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cleared vegetation treatment, hand-weeding was conducted after each monitoring to remove seedlings of spontaneous species whose abundance was generally low (except for the first measurement at the beginning of the growing season) and consistent between sites.

Statistical analysis

For the seed-sowing experiment, the effects of shade level and soil type on the recruitment of *O. sensibilis*, *M. struthiopteris* and *G. striata* were assessed with a two-way ANOVA using the number of sporophytes or seedlings emerging from experimental units at the end of the greenhouse experiment as response variables. For the transplant experiment, the combined effects of the riparian habitat type and microsite (hereafter referred to as treatment, a qualitative variable with six levels) on transplant survival at the end of the second growing season were evaluated for each species by comparing likelihood ratios of nested generalized linear mixed models (GLMM) using block nested in site nested in river as a random variable. For transplant growth, a PCA based on the five morphological measurements was conducted for each species to calculate a size index corresponding to the scores of each transplant along PCA axis 1. This procedure was performed on the morphological measurements taken at the end of the second growing season and those collected the week after transplanting in order to calculate, respectively, the final size index of living transplants and the initial size index of living transplants. For each species, the first axis of the two PCA explained a large proportion of the variation in morphological measurements (between 56% and 75%) and strongly correlated positively with each of the morphological measurements (Appendix S3). These size indices were thus considered representative of transplant size. The effect of treatment on transplant size index at the end of the second growing season were then assessed by comparing likelihood ratios of nested linear mixed models (LMM) using block nested in site nested in river as a random variable and the initial size index of transplants as a co-variable.

Six *a priori* contrasts were performed between treatments to assess the specific effects of riparian habitat type, competition and soil type on the survival and growth of transplants using least square means calculated, respectively, from the GLMM and the LMM mentioned above (Tables 2 and 3). All analyses were conducted on SAS v 9.4 (SAS Institute Inc., Cary, NC, US).

Results

Seedling and sporophyte emergence

Shade level and soil type affected the recruitment of two of the three studied forest herbs (Table 1). Seedling emergence of *G. striata* tended to depend on soil type (Table 1), with fewer seedlings on agricultural compared to forest soil. For *M. struthiopteris*, shade level affected sporophyte emergence (Table 1), with fewer sporophytes emerging under 50% (*PLANTATION* conditions) than under 75% of shade (*FOREST* conditions). Only the germination of *O. sensibilis* was not influenced by shade level or soil type, although insignificantly higher on forest soils and under 75% shade.

Seedling and sporophyte survival

After two growing seasons, the survival of seedlings transplanted in the field was influenced by the treatment effect combining riparian habitat type and microsite for the three studied species, as shown by the GLMM (Table 2, line 'treatment'). Furthermore, *a priori* contrasts showed that transplant survival was similar for the three species between *PLANTATION* and *FOREST*, both in the presence and absence of competing vegetation (Table 2, contrasts 1 and 2). *PLANTATION* thus provided ecological conditions suitable enough to ensure seedling and sporophyte survival similar to that observed in *FOREST*. Nevertheless, for the three species, competition limited transplant survival in *FOREST*, as revealed by the contrasts between vegetation and cleared vegetation in *FOREST* on *in-situ* soil (Table 2, contrast 3). For *G. striata* and *O. sensibilis* this negative effect of competition on transplant survival was also identified in *PLANTATION* (Table 2, contrast 4). The transplant survival of these two species was also reduced by the replacement of forest soil by agricultural soil in *FOREST* (Table 2, contrast 5).

Seedling and sporophyte growth

Transplant size after two growing seasons was influenced for the three studied species by the initial size of the transplant measured the week after transplanting (used as a co-variable), and by the treatment effect (combining riparian habitat type and microsite) for *G. striata* and *M. struthiopteris* (Table 3, line 'treatment'). As for transplant survival, *a priori* contrasts also indicated that the transplant size index of the three species was similar between *FOREST* and *PLANTATION*, both in the presence and absence of vegetation (Table 3, contrasts 1 and 2). The ecological conditions of *PLANTATION* were thus just as suitable as those provided by FOREST for seedling and sporophyte growth of the three studied species. For *G. striata* and *M. struthiopteris*, however, competing vegetation reduced transplant growth in *FOREST* (Table 3, contrast 3). This negative effect of competition on transplant growth was also identified in *PLANTATION* for *G. striata* (Table 3, contrast 4), while forest soil increased the growth

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Table 1. Seedling and sporophytes emergence of the three studied species in response to two shade levels and two soil types, after 14 cultivation weeks in greenhouse, obtained by ANOVA.

		Glyceria striata	*Matteuccia struthiopteris*	*Onoclea sensibilis*
Average Number of Seedlings and Sporophytes (±SE)				
Shade Level	50%	13.6 (±1.7)	143.1 (±15.2)	86.7 (±20.1)
	75%	14.5 (±1.6)	190.8 (±5.9)	106.0 (±18.7)
Soil Type	Agricultural	12.2 (±1.4)	159.1 (±16.8)	78.9 (±21.6)
	Forest	15.9 (±1.7)	147.9 (±7.8)	113.9 (±16.0)
Fixed Effects and Interaction				
	df	*F*-value – *P*-value		
Shade Level	1	0.13 – 0.7302	7.25 – **0.0336**	0.38 – 0.5606
Soil Type	1	4.44 – *0.0568*	1.29 – 0.2777	2.20 – 0.1638
Shade Level × Soil Type	1	0.53 – 0.4796	3.59 – 0.0824	0.00 – 0.9953

P-values inferior to 0.05 are indicated in bold, and *P*-values comprised between 0.05 and 0.1 are in italics.

Table 2. Transplant survival of the three studied species in response to two riparian habitat types and three microsite treatments after two growing seasons.

		Glyceria striata	*Matteuccia struthiopteris*	*Onoclea sensibilis*
Average Transplant Survival (±SE, %)				
Riparian Habitat Type	Microsite			
PLANTATION	*In-situ* soil with vegetation	48.3 (±7.1)	56.7 (±9.3)	38.3 (±10.3)
	In-situ soil with cleared vegetation	70.0 (±8.9)	65.0 (±7.6)	63.3 (±6.8)
	Ex-situ soil with cleared vegetation	81.7 (±7.1)	75.0 (±8.8)	68.3 (±8.6)
FOREST	*In-situ* soil with vegetation	58.3 (±8.3)	61.1 (±10.6)	53.3 (±9.1)
	In-situ soil with cleared vegetation	78.3 (±5.4)	76.7 (±6.9)	71.7 (±9.4)
	Ex-situ soil with cleared vegetation	53.3 (±9.1)	87.8 (±4.9)	51.7 (±9.6)
Fixed Effects				
	df		*F*-value – *P*-value	
Treatment	5	4.94 – **0.0008**	3.76 – **0.0053**	4.29 – **0.0023**
Contrasts				
			F-value – *P*-value	
1) *PLANTATION* vs *FOREST*, *In-situ* Soil with Vegetation		0.58 – 0.4510	0.19 – 0.6677	0.94 – 0.3369
2) *PLANTATION* vs *FOREST*, *In-situ* Soil with Cleared Vegetation		0.63 – 0.4322	1.65 – 0.2041	0.65 – 0.4238
3) Vegetation vs Cleared Vegetation in *FOREST*, *In-situ* Soil		5.80 – **0.0193**	4.14 – **0.0467**	5.65 – **0.0209**
4) Vegetation vs Cleared Vegetation in *PLANTATION*, *In-situ* Soil		6.07 – **0.0168**	1.06 – 0.3082	8.35 – **0.0055**
5) *In-situ* Soil vs *Ex-situ* Soil in *FOREST*, Cleared Vegetation		8.64 – **0.0048**	2.56 – 0.1153	6.61 – **0.0128**
6) *In-situ* Soil vs *Ex-situ* Soil in *PLANTATION*, Cleared Vegetation		2.31 – 0.1339	1.70 – 0.1980	0.38 – 0.5404

P value inferior to 0.05 obtained by generalized linear mixed models and contrasts are indicated in bold.

of transplant *M. struthiopteris* in *PLANTATION* compared to agricultural soil (Table 3, contrast 6). Only the growth of *O. sensiblis* sporophytes was independent of competition and soil type both in *PLANTATION* and *FOREST*.

Discussion

Recruitment is a crucial step for plant recolonization, as the ecological niche space is often narrower for seedling emergence and establishment than for the persistence of adult plants (Grubb 1977; Young et al. 2005). In this study, we showed that planting trees in post-agricultural riparian zones fosters the establishment of forest herbs. Seedling survival and growth of *G. striata*, *M. struthiopteris* and *O. sensiblis* were indeed similar between tree-planted riparian zones and natural riparian forests for microsites on *in-situ* soil, both in the presence and absence of competing vegetation. However, environmental conditions prevailing specifically in riparian zones planted with trees (17 yrs ago) act as additional filters offsetting the positive influence of planted trees on both seedling emergence and establishment (i.e. survival and growth). In tree-planted riparian zones, competition indeed reduced seedling survival for *G. striata* and *O. sensibilis* as well as seedling growth for *G. striata*, while agricultural soils reduced seedling growth for *M. struthiopteris*. Furthermore, additional environmental limitations were identified during seedling emergence for two of the three studied species: *G. striata*

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Table 3. Transplant size index of the three studied species in response to two riparian habitat types and three microsite treatments after two growing seasons.

		Glyceria striata	*Matteuccia struthiopteris*	*Onoclea sensibilis*
Average Transplant Size Index (±SE)				
Riparian Habitat Type	Microsite			
PLANTATION	*In-situ* soil with vegetation	−0.12 (±0.03)	−0.08 (±0.04)	−0.11 (±0.06)
	In-situ soil with cleared vegetation	0.15 (±0.05)	0.02 (±0.04)	−0.01 (±0.06)
	Ex-situ soil with cleared vegetation	0.04 (±0.03)	0.12 (±0.05)	0.04 (±0.05)
FOREST	*In-situ* soil with vegetation	−0.16 (±0.03)	−0.12 (±0.04)	0.02 (±0.06)
	In-situ soil with cleared vegetation	−0.04 (±0.05)	0.01 (±0.06)	0.03 (±0.06)
	Ex-situ soil with cleared vegetation	0.09 (±0.06)	0.01 (±0.05)	−0.03 (±0.07)
Fixed Effects	df		*F* value – *P* value	
Initial Size Index	1	10.36 – **0.0015**	16.21 – **<0.0001**	33.55 – **<0.0001**
Treatment	5	10.45 – **<0.0001**	9.15 – **<0.0001**	1.63 – 0.1541
Contrasts			*F* value – *P* value	
1) *PLANTATION* vs *FOREST*, *In-situ* Soil with Vegetation		2.18 – 0.2004	1.03 – 0.3102	<0.01 – 0.9756
2) *PLANTATION* vs *FOREST*, *In-situ* soil with Cleared Vegetation		2.64 – 0.1288	0.01 – 0.9107	<0.01- 0.9963
3) Vegetation vs Cleared Vegetation in *FOREST*, *In-situ* Soil		**0.0001**	14.33 – **0.0002**	2.52 – 0.1142
4) Vegetation vs Cleared Vegetation in *PLANTATION*, *In-situ* Soil		17.87 – **<0.0001**	0.94 – 0.3328	1.97 – 0.1619
5) *In-situ* Soil vs *Ex-situ* Soil in *FOREST*, Cleared Vegetation		1.01 – 0.3177	0.94 – 0.3328	2.98 – 0.0863
6) *In-situ* Soil vs *Ex-situ* soil in *PLANTATION*, Cleared Vegetation		0.65 – 0.4203	7.70 – **0.0060**	0.33 – 0.5690

P value inferior to 0.05 obtained by generalized linear mixed models and contrasts (using transplant initial size index as a co-variable) are indicated in bold. The plant size index corresponds to the scores of transplants along the first axis of a PCA based on the different morphological measurements for each species. High size index correspond to tall and large transplant with numerous stems and leaves.

and *M. struthiopteris* emergence was, respectively, reduced by agricultural soil relative to forest soil and by low shade levels typical of tree-planted riparian zones. These complementary results highlight partial success of tree planting to restore the establishment niches of forest herbs, as their recolonization is still limited in tree-planted riparian zones. Cumulative effects of environmental filters acting at the seedling emergence and/or establishment stages thus appear partly responsible for the absence of forest herbs in tree-planted riparian zones. In recent forests growing in post-agricultural landscapes, the assembly rules of plant communities were mainly related to a combined effect of dispersal abilities and recruitment and establishment requirements (Clark et al. 2007; Hermy & Verheyen 2007). Our study demonstrates that competitive abilities and resource-use strategies indeed strongly influence species recolonization and community composition in tree-planted riparian zones.

Environmental limitations similar to those observed here in tree-planted riparian zones have already been identified in recent forests where competition usually, combined with soil P availability, reduced the germination of forest herbs (De Keersmaeker et al. 2004; Verheyen & Hermy 2004; Baeten et al. 2009a; Orczewska 2009). As for *G. striata* and *O. sensiblis,* competition similarly reduces seedling establishment and adult growth of the two forest herbs *Primula elatior* (L.) Hill. and *Geum urbanum* L. (Verheyen & Hermy 2004) in planted recent mesic forests developing after stopping agricultural activities. In addition, the higher soil nutrient content in recent forests relative to ancient forests results in higher adult growth for forest hemicryptophytes, but not geophytes (Verheyen & Hermy 2004). In our study, agricultural soils had a detrimental effect on sporophyte growth of the geophyte fern *M. struthiopteris*. While soil P content was here similar between tree-planted riparian zones and natural riparian forests, in contrast to the above-mentioned studies, it is likely that the higher proportion of organic matter and soil RH of forest soils fostered forest herb establishment by preventing water scarcity and temporary drought, and creating suitable microsites for recruitment. Furthermore, light availability, and more precisely low shade levels, also reduced the emergence of *M. struthiopteris* and has already been proven as a major abiotic factor limiting growth of forest herbs in mesic to eutrophic recent forests (Baeten et al. 2010). Prolonged high light availability is thus an important filter limiting forest herb recolonization, even if momentary increase in light levels related to forest management or natural dynamics can favour such species in mesic forests (De Keersmaeker et al. 2011; Thomaes et al. 2011). Our study therefore showed that the species assembly of tree-planted riparian zones is conditioned by ecological filters related to soil conditions and light availability, as previously identified in mesic recent forests.

Compared to mesic forests, riparian plant communities are, however, strongly dependent on hydrological regimes and disturbances, as water flow regularly reorganizes soils in riparian zones through erosion and deposition processes and modifies competitive interactions through the destruction of plant communities (Hupp & Osterkamp 1996; Tabacchi et al. 1998; Bendix & Hupp 2000; Osterkamp & Hupp 2010). Recent studies based on transplant experiments have, for example, highlighted that riparian herbaceous species are characterized by increased seedling survival and growth in rivers with restored flow regimes compared to channelized rivers (Dietrich et al. 2015, 2016). In our study, hydrological dynamism appeared too low to mitigate the legacies of agricultural land use on soils and plant competition or foster the recolonization of forest herbs, although long-term experiments are required to fully assess the persistence of past land-use impacts.

Our study clearly identified the existence of species-specific ecological requirements for the recolonization of forests herbs in tree-planted riparian zones, similarly to those observed in recent mesic forests (Verheyen & Hermy 2004; Baeten et al. 2009a,b). In addition, we showed that two of the studied species were limited by distinct environmental filters, depending on their development stage. Indeed, the emergence of *G. striata* seedlings was limited by agricultural soils, but not its seedling survival and growth, which was rather limited by competition in tree-planted riparian zones. For *M. struthiopteris*, low shade levels reduced seedling emergence, but not agricultural soils, which however limited seedling growth. These specific ecological requirements related to life-cycle stages evidence ontogenic niche shifts (Young et al. 2005) between seeds/spores and seedlings, which cumulatively reduce recruitment and strongly limit recolonization.

Determining species-specific requirements for recruitment and establishment by combining seed-sowing and transplant experiments is an approach of great interest for ecological restoration, despite the high costs and effort required for such experiments, both during implementation and long-term monitoring. Although only three species were studied here, it seems that controlling spontaneous vegetation in the first years after tree planting is advisable for fostering the recolonization of forest herbs that experience competitive limitation, such as *G. striata* and *O. sensibilis*. Given the effect of soil condition on species recruitment, assessing the growth of forest herbs on contrasting soils with different chemical composition (especially nutrient content), physical properties (such as compaction) or microbial communities is also advised to identify the specific physico-chemical parameters that reduce recolonization. Future experiments should also aim to quantify forest herb recruitment under distinct tree species in order to identify potential species-specific overstorey–understorey interactions detrimental to restoration. As previously pointed out (Dietrich et al. 2013), the precise identification of ecological filters that reduce plant recolonization using transplant experiments conducted over the long term rather than the short term (notably to avoid lag effects in ecosystem response) is promising and should be more generally considered to refine restoration strategies and improve restoration success.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Environmental variables (mean ± standard error) characterizing the five natural riparian forests and the five tree-planted riparian zones selected among the two agricultural watersheds to perform the transplant experiment.
Appendix S2. Plant species inventoried in the ten sites selected for the field transplant experiment (mean cover in the three microsites on *in-situ* soil with vegetation, in %).
Appendix S3. Variation explained by the first axis of the PCA used to calculate for each of the three studied species the transplant size index the week after transplanting (initial size) and the transplant size index after two growing seasons, and R-square of the linear correlations between each morphological measurement and the transplant scores along the first PCA axis (corresponding to the transplant size index).

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